

LATERRIERE, QC, Canada

WMO: 710400

Lat: 48.3072N

Lon: 71.1290W

Elev: 163

StdP: 99.39

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.9	-27.4	-34.3	0.2	-29.5	-31.6	0.2	-27.1	10.5	-13.1	9.7	-13.2	2.1	220	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.8	19.5	26.9	19.0	25.2	18.2	21.4	26.0	20.3	24.9	19.3	23.7	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.8	23.9	18.6	13.7	22.7	17.6	12.8	21.7	62.9	26.2	59.2	24.9	55.7	23.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.3	7.3	-34.1	32.0	2.9	1.3	-36.2	32.9	-37.9	33.7	-39.5	34.4	-41.7	35.4
			-34.2	23.4	3.0	1.1	-36.4	24.2	-38.1	24.8	-39.7	25.4	-41.9	26.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-14.8	-13.1	-6.7	2.0	10.0	15.2	18.2	17.2	12.8	5.9	-1.3	-9.8
	DBStd	12.65	7.92	7.00	6.68	4.70	4.25	3.63	2.90	3.00	4.14	4.28	5.13	6.84
	HDD10.0	3343	769	646	519	243	53	2	0	0	17	142	339	613
	HDD18.3	5656	1027	879	777	491	261	107	37	57	173	387	589	871
	CDD10.0	808	0	0	1	1	53	159	256	223	100	14	1	0
	CDD18.3	79	0	0	0	0	2	13	34	22	7	1	0	0
	CDH23.3	827	0	0	1	1	61	192	326	179	65	2	0	0
	CDH26.7	177	0	0	0	0	16	49	71	32	9	0	0	0
Wind	WSAvg	3.2	3.4	3.5	3.6	3.7	3.3	3.2	2.9	2.9	2.7	3.0	3.3	3.1
Precipitation	PrecAvg	958	62	53	57	59	80	91	116	101	102	85	77	75
	PrecMax	1220	122	112	128	106	181	160	320	245	226	149	147	142
	PrecMin	754	17	13	12	12	15	19	37	35	41	27	22	27
	PrecStd	105	24	23	26	25	33	32	45	41	36	29	26	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	3.6	13.7	20.2	29.2	30.9	30.9	29.7	27.7	22.5	14.8	6.2
		MCWB	4.3	1.5	8.8	10.7	16.8	20.5	20.7	20.9	19.9	16.2	10.9	4.4
	2%	DB	2.4	1.3	6.8	15.2	24.7	28.0	28.6	27.3	25.3	19.0	9.9	2.6
		MCWB	1.5	-0.1	2.8	7.5	14.8	18.0	20.0	19.8	19.3	14.0	7.0	1.5
	5%	DB	0.1	-0.1	4.5	11.8	21.6	25.4	26.8	25.2	22.9	15.5	7.4	1.1
		MCWB	-0.7	-1.3	1.4	6.1	13.0	17.3	19.2	18.9	17.7	11.5	5.4	0.2
	10%	DB	-2.8	-2.0	2.6	9.0	18.8	23.2	25.0	23.5	20.5	12.6	5.3	-0.1
		MCWB	-3.6	-3.1	0.5	4.6	12.2	16.1	18.6	18.1	16.3	9.8	3.9	-0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	1.9	9.2	12.3	19.4	22.1	23.1	22.2	21.7	16.9	11.4	4.3
		MCDB	5.9	3.1	11.7	18.3	24.5	28.6	28.0	27.0	26.0	21.1	14.6	5.8
	2%	WB	1.7	0.2	3.5	8.5	16.8	20.1	21.4	20.8	19.9	14.4	7.8	1.7
		MCDB	2.4	1.2	5.9	14.1	23.3	25.3	25.9	25.4	23.9	18.4	9.3	2.4
	5%	WB	-0.6	-1.4	1.9	6.7	14.2	18.7	20.4	19.7	18.3	12.2	5.6	0.4
		MCDB	0.2	-0.2	4.0	10.8	19.8	23.9	24.8	24.2	22.3	14.9	7.0	1.1
	10%	WB	-3.8	-3.1	0.5	5.2	12.5	17.2	19.2	18.8	16.8	10.1	4.0	-0.9
		MCDB	-2.9	-1.9	2.3	8.5	18.0	21.8	23.6	22.6	20.2	12.3	5.5	0.0
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.7	11.0	10.0	12.3	11.7	10.9	10.5	11.0	8.3	7.2	8.2
		MCDBR	11.6	9.8	13.4	15.6	18.0	16.3	14.3	14.0	14.7	13.7	10.2	8.0
		MCWBR	10.7	8.7	10.1	9.8	9.3	8.5	7.2	7.6	9.0	9.1	8.0	7.2
	5% WB	MCDBR	11.3	9.1	10.8	13.5	14.9	13.6	11.9	11.8	13.0	12.7	9.0	8.0
		MCWBR	10.6	8.2	8.5	9.0	8.9	8.1	6.9	6.8	8.6	9.1	7.4	7.5
Clear-Sky Solar Irradiance	taub	0.266	0.284	0.315	0.376	0.358	0.383	0.402	0.381	0.370	0.329	0.316	0.266	
	taud	2.266	2.224	2.201	2.222	2.408	2.367	2.326	2.387	2.407	2.505	2.374	2.322	
	Ebn at Noon	804	870	897	871	903	879	855	859	833	820	735	758	
	Edh at Noon	74	97	118	130	112	118	121	109	97	75	68	62	
All-Sky Solar Radiation	RadAvg	1.31	2.27	3.51	4.45	5.15	5.49	5.20	4.43	3.40	1.95	1.07	0.87	
	RadStd	0.16	0.30	0.29	0.35	0.50	0.37	0.51	0.29	0.26	0.20	0.11	0.09	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page